The Invariance Principle — problem 1.44

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Can you get the polynomial $h(x) = x$ from the polynomials $f(x)$ and $g(x)$ by the operations **addition, subtraction, multiplication** if

(a) $f(x) = x^2 + x$, $g(x) = x^2 + 2$; (b) $f(x) = 2x^2 + x$, $g(x) = 2x$;

(c) $f(x) = x^2 + x$, $g(x) = x^2 - 2$?